

ສູນວິສະວະກຳເພື່ອການພັດທະນາແບບຍືນຍົງ

“The Center of Engineering for Sustainable Development (CESD)”



ມະຫາວິທະຍາໄລ ແຫ່ງຊາດ
National University of Laos

ຄະນະ ວິສະວະກຳສາດ
Faculty of Engineering

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2. About CESD

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Background

- The Japanese government's grant a **Project for Strengthening Human Resource Development of Engineering and Technology for Industry Development in Lao PDR.**
- Total Grant: 2.5 billion yen, was handover on 05/05/2025.
- The project includes the construction of 1 laboratory building and 1 Engineering Center for Sustainable Development



Concept

The JICA Grand Aid Project considers the following concept for the CESD buildings and equipment as followed:

- **Collaboration research/development** using the equipment designated for CESD
- **Product inspections and performance tests by enterprises**, using the CESD equipment
- **Measurement and analysis services** provided as a social service of FEN for external organizations
- **Facilitation of academic-industrial collaboration** using meeting rooms in CESD
- **Appeal the achievements of academic-industrial collaboration** activities using the exhibition room

2. About CESD

Objective of the CESD

The Objective: **A Hub for Innovation**, The Center of Engineering for Sustainable Development is the premier academic and research hub at the Faculty of Engineering (FEN), NUOL. “We are dedicated to bridging the gap between theoretical knowledge and practical application”

Our Goal: To foster the next generation of engineers and provide **a platform for sustainable industrial growth**.

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3. The Vision and Mission

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The Center of Engineering for Sustainable Development's Vision and Mission (2025-2030)

ວິໄສທັດ | VISION

To be recognized as the leading national center for academic-industrial collaboration and a key driver of sustainable development within the engineering sector of Lao PDR. We aspire to be a beacon of innovation, quality, and practical expertise.

ພັນທະກິດ | MISSION

- To serve as a central nexus for fostering powerful collaborations between academia and industry.
- To elevate the quality of engineering education and research at FEN-NUOL to international standards.
- To provide state-of-the-art facilities and invaluable resources that are accessible to all departments within FEN-NUOL.

6. Laboratories under CESD and Research Facilities

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CESD : Center of Engineering For Sustainable Development

- Chemical Analysis Laboratory
- Electron Microscope Laboratory
- X-ray Analysis Laboratory
- High-Voltage Laboratory
- Solid Forming Laboratory
- Computer Room
- Physical Measurement Laboratory
- Electrical and Electronic Measurement Laboratory
- **Admin & Common**
 - Meeting Room
 - Exhibition Room
 - Administration Office
 - etc.



7. Services provide by the CESD:

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Services for Education:

- **Student Project Support:** Providing essential equipment for undergraduate and postgraduate research.
- **Curriculum Integration:** Integrating hands-on experimental work into the core engineering curriculum.
- **Faculty Research:** Supporting faculty research projects by providing access to advanced testing and analysis tools



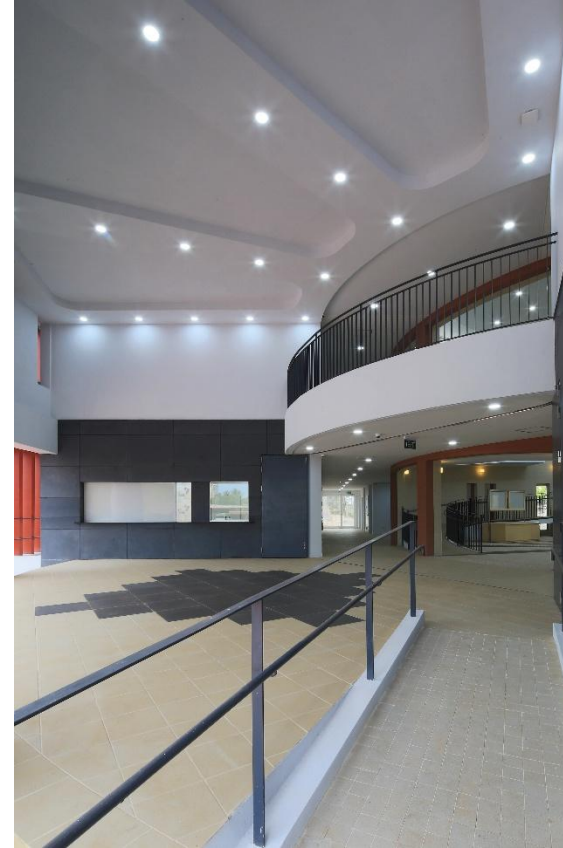
7. Services provide by the CESD:

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Services for Research:

- **Contracted Analysis:** Offer measurement and analysis services to external companies, providing reliable, third-party data.
- **Joint R&D:** Partner on collaborative research projects to solve real-world industrial challenges.
- **Talent Development:** Serve as a direct pipeline for industry recruitment and practical training.



7. Partner with the CESD:

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Partner with Us: Industry

- **Access R&D and Specialized Analysis:**

Utilize our state-of-the-art laboratory facilities and collaborate on joint R&D projects.

- **Build Your Talent Pipeline:** Connect with our top engineering students and faculty, serving as a direct pipeline for recruitment and project support.

- **Develop Custom Training:** Create customized training programs and workshops for your employees, leveraging our academic expertise



7. Partner with the CESD:

Partner with Us: Academia

- **Collaborative Research:** Engage in joint research projects with our faculty and access our specialized facilities.
- **Knowledge Exchange:** Partner on workshops, seminars, and exchange programs to share expertise and resources.
- **Facility Sharing:** Explore opportunities to use our advanced laboratories and equipment for your own research needs.



8. Conclusions

- The CESD has successfully integrated advanced equipment into the FEN curriculum and provided a platform for numerous student and faculty projects, fostering a new generation of skilled engineers.
- Looking ahead, our goals include expanding our research focus to new areas of sustainable technology, securing additional industry partnerships, and establishing CESD as a regional leader in engineering education



ຂອບໃຈ
Thank you

Let's Engineer a Sustainable
Future, Together.

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